

吴筱琦

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现居城市：伦敦/苏州

意向城市：上海/苏州/杭州 | 生物医药研发/QC/分析岗



教育背景

帝国理工学院 (QS 2) 计算生命科学研究型硕士

伦敦 2025.09 - 2026.09

帝国理工学院 (QS 2) 生物科学本科

伦敦 2022.09 - 2025.07

- 相关课程：应用分子生物学，遗传学，细胞生物学与遗传学，生物化学与微生物学，生物信息学，编程与统计

实习经历

上海宣泰医药科技有限公司 分析工程师 · 分析部

上海 2024.07 - 2024.09

- 稳定性检测：使用高效液相色谱-紫外检测 (HPLC/UV) 技术对客户提供的药品样本进行稳定性检测，专注于样品中杂质的检测与分析
- 数据分析：独立完成样本前处理、HPLC 系统操作及数据分析，确保检测结果的准确性和可靠性
- 质量控制：通过严格的质控流程鉴别潜在杂质，为药品的稳定性评估提供关键数据支持

科研与项目经历

多模态大语言模型提取媒介性状数据评估：高精确率与结构依赖型记录召回

伦敦 2025.12 - 2026.08

Python、Gemma 4 31B IT (多模态 LLM)、PyMuPDF、Docling、提示工程、JSON Schema 校验、pytest

- 基准构建与提取流程：设计加权多样性贪心算法从 VecTraits 超 4 万行、334 篇论文中选取 20 篇，构建 2,335 条开发基准与 431 条留出测试集；结合 PyMuPDF/Docling 证据提取，驱动 Gemma 4 31B IT 完成提示工程约束下的可溯源性状记录提取
- 严谨评估与质量验证：设计二分图记录匹配评估框架，在留出集上验证锁定流程，取得记录精确率 1.000、条件宏观字段 F1 0.729，体现与药品质量控制一致的系统性数据验证思维

病毒指纹图谱：利用 HERV-K solo-LTR 多态性进行人群遗传学分析

伦敦 2023.11 - 2023.12

PCR、凝胶电泳、群体遗传学分析

- 实验操作：运用 PCR 和凝胶电泳检测 5 组引物对应的基因组多态性，结合 F_{ST} (范围 0.0010-0.3659) 和 HWE (χ^2 检验，临界值 3.841) 进行群体遗传学分析
- 分析评估：评估 HERV-K solo-LTR 作为人群遗传与法医学标记的潜力与挑战，识别显著偏离 HWE 的引物位点
- 研究成果：完成实验设计、操作与数据分析全流程，成果获优秀评价

OCA1A 型白化病的生物信息学分析

伦敦 2023.10 - 2023.11

NCBI、OMIM、Ensembl、UniProt、AlphaFold、ClinVar

- 结构分析：分析 TYR 基因突变对酪氨酸酶结构与功能的影响，聚焦铜结合域突变，并使用 AlphaFold 结构预测辅助分析
- 数据库整合：跨 NCBI、OMIM、Ensembl、UniProt、ClinVar 数据库进行变异注释与人群频率分析
- 研究成果：完成生物信息学报告，掌握多数据库整合、蛋白结构预测与进化保守性分析技能

近交系小鼠在医学研究中的应用与局限性综述

伦敦 2023.11 - 2023.12

文献综述、遗传学分析

- 证据综合与批判分析：阐述近交系小鼠遗传学原理及其临床前药物应用，并批判分析其遗传/生理差异等局限性，成果获优秀评价

技能/其他

- 分子与生化实验：PCR 与凝胶电泳、DNA/RNA 提取 (Qiagen 试剂盒)、蛋白质电泳、西方印迹；群体遗传学分析 (F_{ST} 、Hardy-Weinberg 平衡/ χ^2 检验)；跨 NCBI、OMIM、Ensembl、UniProt、ClinVar、AlphaFold 的变异注释与蛋白结构分析
- 数据分析：R (ggplot2、dplyr)、Python、统计推断 (GLM、Wilcoxon 检验、t 检验、Pearson 相关)、质量控制文档规范
- 语言：英语 (雅思 7.5)，普通话 (母语)，法语 (A2)；兴趣爱好：钢琴，滑雪

Xiaoqi Wu

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Currently based in: London / Suzhou

Biopharma R&D | QC | Analytical | Shanghai / Suzhou / Hangzhou

Education

Imperial College London MSc Computational Methods in Ecology and Evolution London 2025.09 - 2026.09

Imperial College London BSc Biological Sciences London 2022.09 - 2025.07

- **Relevant Coursework:** Applied Molecular Biology, Genetics, Cell Biology and Genetics, Biochemistry and Microbiology, Bioinformatics, Programming and Statistics

Internship Experience

Shanghai Sunherb Pharmaceutical Technology Co., Ltd. Analytical Shanghai 2024.07 - 2024.09
Engineer · Analytical Department

- **Stability Testing:** Conducted stability testing of client-supplied drug samples using HPLC/UV, focusing on impurity detection and analysis
- **Data Analysis:** Independently carried out sample pre-treatment, HPLC system operation, and data analysis, ensuring accuracy and reliability of test results
- **Quality Control:** Identified potential impurities through rigorous quality-control procedures, providing key data to support drug stability assessments – demonstrating solid technical skills in pharmaceutical analysis and instrument operation

Research & Project Experience

Evaluating Multimodal LLM Extraction for Vector Trait Curation: High Precision with Structure-Dependent Record Recovery London 2025.12 - 2026.08

Python, Gemma 4 31B IT (multimodal LLM), PyMuPDF, Docling, prompt engineering, JSON Schema validation, pytest

- **Benchmark Construction:** Designed a weighted greedy diversity-selection algorithm to select 20 papers from VecTraits' 40,000+-row, 334-paper raw export; built a 2,335-record manually curated development benchmark and a 431-record held-out test set, reduced from VecTraits' 91-column schema to a 22-field extraction schema aligned with the MIREVTD reporting standard
- **Multimodal Extraction Pipeline:** Combined PyMuPDF text extraction and Docling table-structure parsing to drive Gemma 4 31B IT (a 256K-context multimodal LLM) through 11 iterated prompt versions, reaching a development-set conditional macro field F1 of 0.810
- **Rigorous Evaluation & Quality Verification:** Designed a bipartite record-matching evaluation framework and validated the locked pipeline on the held-out test set, achieving row precision of 1.000 and conditional macro field F1 of 0.729 – reflecting the same rigorous, systematic data-validation mindset applied in pharmaceutical quality control

Viral Fingerprinting: Population Genetics Analysis Using HERV-K solo-LTR Polymorphism London 2023.11 - 2023.12

PCR, gel electrophoresis, population genetics analysis

- **Lab Work:** Detected genomic polymorphisms across 5 primer sets using PCR and gel electrophoresis, combined with F_{ST} (range 0.0010-0.3659) and HWE (χ^2 test, critical value 3.841) for population genetics analysis
- **Analysis:** Evaluated the potential and challenges of HERV-K solo-LTR as a population genetic and forensic marker, identifying primer sites significantly deviating from HWE
- **Outcome:** Completed the full experimental design, lab work, and data analysis workflow; results received distinction

Bioinformatic Analysis of OCA1A Albinism

London 2023.10 - 2023.11

NCBI, OMIM, Ensembl, UniProt, AlphaFold, ClinVar

- **Structural Analysis:** Analyzed the effect of TYR gene mutations on tyrosinase structure and function, focusing on copper-binding domain mutations, using AlphaFold structural predictions

- **Database Integration:** Performed variant annotation and population-frequency analysis across NCBI, OMIM, Ensembl, UniProt, and ClinVar, covering population-frequency data for 85,874 samples
- **Outcome:** Completed a bioinformatics report, developing skills in multi-database integration, protein structure prediction, and evolutionary conservation analysis

Review: Applications and Limitations of Inbred Mouse Strains in Medical Research London 2023.11 - 2023.12

Literature review, genetic analysis

- **Evidence Synthesis:** Systematically described the genetic principles of inbred mouse strains (>98% allelic homozygosity after 20+ generations) and their use in preclinical drug research, e.g. 40 BALB/c mice evaluating a melittin-loaded nanoliposome formulation against free melittin for breast cancer (renal/liver enzyme assays)
- **Critical Analysis:** Critically analyzed the inbred-mouse model's limitations – genetic/physiological differences from humans, inbreeding depression, and ethical concerns; review received distinction

Skills & Other

- **Molecular & Biochemical Techniques:** PCR and gel electrophoresis (multi-primer genotyping, e.g. 5-primer-set solo-LTR detection), DNA/RNA extraction (Qiagen kits), protein electrophoresis, Western blotting; population genetics analysis (F_{ST} , Hardy-Weinberg equilibrium/ χ^2 tests); variant annotation & protein structure analysis across NCBI, OMIM, Ensembl, UniProt, ClinVar, and AlphaFold
- **Data Analysis:** R (ggplot2, dplyr), Python, statistical inference (GLM, Wilcoxon signed-rank, t-tests, Pearson correlation), quality-control documentation standards
- **Languages:** English (IELTS 7.5), Mandarin Chinese (native), French (A2) | **Interests:** piano, skiing